January 14th
To-dos:

- Confirm you can access Lugh HPC system
  - Anthony's Lugh guide: https://maths.universityofgalway.ie/computecluster/ (Notes things we talked about like appropriate use of nodes, queue scheduling, etc.)
- Create a new GitHub repository that we will use for this project
  - cnv-analysis-toolkit
  - CAT-Kit
  - Add my GitHub account to the repo: https://github.com/GWheeler01
- Read the GATK CNV pipeline documentation and attempt the tutorial: https://gatk.broadinstitute.org/hc/en-us/articles/360035531092--How-to-part-I-Sensitively-detect-copy-ratio-alterations-and-allelic-segments
  - The tutorial *may* be doable on a MacBook or using WSL; you will definitely need Lugh to run real data.
  - I am working on getting the actual data we will need to make sure this works on real samples, but the linked guide should provide minimal test inputs for you to use.
  - I recommend downloading the GATK4 Java executable instead of their Docker container (as otherwise you'd have to put *their* Docker container inside of *your* Docker container later on)
- Try your hand at making a flowchart describing a software workflow.
  - Could be good to illustrate the above GATK pipeline from the tutorial as an exercise.
  - For the full project, will need to include all possible inputs and the necessary steps needed to get them into the CNV UI's input JSON format (we will talk about this more)
  - https://app.diagrams.net/

January 20th

`Username: mcollins`

`Password: tiegh3spsm`

Cluster documentation is available here https://maths.universityofgalway.ie/computecluster/

Use your Home folder for small files, keep this directory under 20GB in size: `/home/username`

Use this area for large data storage: `/data4/mcollins`

January 28th

First, create a file in your home directory called ".condarc". You can create this file with "nano ~/.condarc" or the text editor of your choice. Enter all of the following:

```
channels:
  - bioconda
  - conda-forge
  - r
  - nodefaults
channel_priority: strict
```

… and save. Then, log out of Lugh and back in. What this does is it tells Anaconda that, whenever you make an environment, these are the channels you want it to use for packages. These packages should cover most of the major things we use for bioinformatics, and notably excludes the official "default", which I had mentioned before. (Default can cost the university money, and so I think it has been IP-blocked by Anthony, which means it will produce errors.)

Once you've done this, you can load anaconda with 'module load Anaconda3'. Then, you want to create your base conda environment that you will install general-use software to. I called mine 'main', but you can call yours whatever you want, and it's fine to make several of them for different purposes.

Create the environment with 'conda create -n main'. You can also do 'conda create -n main python==3.10' if for whatever reason, you need to specify a specific older version of Python to use. (Sometimes you may find yourself wanting to use a specific piece of software that hasn't been updated and has become incompatible with newer versions.) Once this is set up, you can activate the environment with 'conda activate main' and then use it as normal.

Below is my .bashrc file. This is the file of commands that will run every time you start your system up. I recommend creating one of these the same way you created the .condarc file, except named '.bashrc'

```
# .bashrc

# Source global definitions
if [ -f /etc/bashrc ]; then
. /etc/bashrc
fi

# User specific aliases and functions
export PATH=$PATH:/data4/wheeler_shared/bin/links/
module load Anaconda3
conda activate main
```

A few notes on this: 'export' sends the text to the right of '=' to the system variable 'PATH'. PATH is a special variable that tells the system where to look for installed software, expecting a colon-separated list of directories. What this does is it adds my shared binaries directory, /data4/wheeler_shared/bin/links to your existing path list, $PATH. This means any binaries I link in there will be runnable for you without

additional installation, which is handy. After this, the 'module load Anaconda3' and 'conda activate main' will make sure that conda is running and in your correct environment when you log in.

After you create this .bashrc file, log out and back in. If you can type 'gatk' and have it run GATK, you have probably done this correctly. To be sure, you can enter 'which gatk' and it will tell you where your GATK executable is located. It should say: /data4/wheeler_shared/bin/links/gatk Finally, as you may have gathered from this, the shared folder for us has been created in **/data4/wheeler_shared**. Currently, you should have read and execute access to this folder, meaning that you can access data and run scripts that I put in there, but would not be able to delete the files or put your own data in here. You will run any scripts you like in your own directories (which are set up for large file storage) and if there are things that should be shared or retained long term, let me know and I will copy them to the shared folder. When you want to run an analysis using files in the shared directory, you should just use the *path* to the location of the files. Do not *copy* the files, as many are very large and may have various security restrictions on them.

In the shared folder so far, you will find the following:

bin : Executables that get added to your path (currently just GATK 4.6.2)
data_files : Data files for various cancers, etc. that you will use for your projects.*
reference : Human reference genome files that you will use for alignment, variant calling, and so on.**

'* : Some data files will be protected access, where I will have had to sign a license agreement that the data will not be used for anything outside of the agreed parameters (under penalty of revocation of any future access, lawsuits, and/or death). I will include an 'ACCESS.txt' file that tells you what the limitations of use are. For example, in the TXCRB folder:

The data files in this directory are from an open-access plot of the Texas Cancer Research Biobank.

They are broadly consented and can be accessed without specific use agreements. The only limitation is that you
CANNOT make any attempt to reidentify the patients.

You may freely use these data to test any analyses or try out new ideas you may have.

https://www.nature.com/articles/sdata201610
(The linked article explains the data and there's also a manifest file with information on the individual samples)

I will soon add a folder that will be labeled Aurora_US, which will be *tightly* controlled data that can only be used for medical research. Once I have downloaded that data, you should not under any circumstances use it for things that fall outside the scope of your projects and you absolutely cannot share any of it with others. If I am able to add additional data, similar restrictions will likely apply. (When you've read this, please reply and confirm that you understand and agree.)

'** : Reference files are things like the Human Reference Genome v38 that will be reused over and over and over again across analyses. It is preferable to keep files like this in a central location, so that we're all using the same versions and not downloading and storing a mess of data unnecessarily. If you've used reference file A for alignment, you want to make sure other analyses also use the exact same version of reference file A, otherwise things will get weird or just break. If there are reference files that you need and you're unable to find them, let me know and I will retrieve them and put them here. If you do download files and you think they should be placed here as a shared resource, let me know.

GATK tutorial through lugh

salloc -n1 -N1 -p MSC

```
(main) [mcollins@lugh ~]$ squeue -u mcollins
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
          10980265       MSC     bash mcollins  R       1:24      1 compute01
(main) [mcollins@lugh ~]$ ssh compute01
Warning: Permanently added 'compute01,192.168.11.11' (ECDSA) to the list of known hosts.
(main) [mcollins@compute01 ~]$ module li

Currently Loaded Modules:
  1) autotools   3) gnu7/7.3.0       5) ohpc                  7) java/22.0.1
  2) prun/1.3    4) openmpi3/3.1.0   6) Anaconda3/2024.02-1
```

```
(main) [mcollins@compute01 ~]$ gatk PreprocessIntervals -L targets_C.interval_list -R
/data4/wheeler_shared/reference/Broad/Homo_sapiens_assembly38_noALT_noHLA_noDecoy.fasta.gz --bin-length 0 --interval-merging-rule
OVERLAPPING_ONLY -O sandbox/targets_C.preprocessed.interval_list
A USER ERROR has occurred: Badly formed genome unclippedLoc: Query interval "targets_C.interval_list" is not valid for this input.
```

Removed -L and tried it again

```
(main) [mcollins@compute01 ~]$ gatk PreprocessIntervals -R
/data4/wheeler_shared/reference/Broad/Homo_sapiens_assembly38_noALT_noHLA_noDecoy.fasta.gz --bin-length 0 --interval-merging-rule
OVERLAPPING_ONLY -O sandbox/targets_C.preprocessed.interval_list

(main) [mcollins@compute01 ~]$ gatk CollectReadCounts -I /data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/tumor.bam -L
sandbox/targets_C.preprocessed.interval_list --interval-merging-rule OVERLAPPING_ONLY -O sandbox/tumor.counts.hdf5

gatk --java-options "-Xmx6500m" CreateReadCountPanelOfNormals \
    -I /data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/HG00133.alt_bwamem_GRCh38DH.20150826.GBR.exome.counts.hdf5 \
    -I /data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/HG00733.alt_bwamem_GRCh38DH.20150826.PUR.exome.counts.hdf5 \
    -I /data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/NA19654.alt_bwamem_GRCh38DH.20150826.MXL.exome.counts.hdf5 \
    --minimum-interval-median-percentile 5.0 \
    -O sandbox/cnvponC.pon.hdf5

gatk --java-options "-Xmx12g" DenoiseReadCounts \
    -I /data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/hcc1143_T_clean.counts.hdf5 \
    --count-panel-of-normals /data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/cnvponC.pon.hdf5 \
    --standardized-copy-ratios sandbox/hcc1143_T_clean.standardizedCR.tsv \
```

```
    --denoised-copy-ratios sandbox/hcc1143_T_clean.denoisedCR.tsv

(main) [mcollins@compute01 sandbox]$ ls
cnvponC.pon.hdf5  hcc1143_T_clean.denoisedCR.tsv  hcc1143_T_clean.standardizedCR.tsv  targets_C.preprocessed.interval_list
tumor.counts.hdf5
(main) [mcollins@compute01 sandbox]$ mkdir plots

gatk PlotDenoisedCopyRatios --standardized-copy-ratios sandbox/hcc1143_T_clean.standardizedCR.tsv --denoised-copy-ratios
sandbox/hcc1143_T_clean.denoisedCR.tsv --sequence-dictionary
/data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/Homo_sapiens_assembly38.dict --minimum-contig-length 46709983 --output
sandbox/plots --output-prefix hcc1143_T_clean

A USER ERROR has occurred: Unable to execute Rscript command: Please add the Rscript directory to your environment ${PATH}
```

Tried to load R

```
(main) [mcollins@compute01 ~]$ module load R
(main) [mcollins@compute01 ~]$ which Rscript
/opt/ohpc/pub/libs/gnu7/R/3.5.0/bin/Rscript

gatk PlotDenoisedCopyRatios --standardized-copy-ratios sandbox/hcc1143_T_clean.standardizedCR.tsv --denoised-copy-ratios
sandbox/hcc1143_T_clean.denoisedCR.tsv --sequence-dictionary
/data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/Homo_sapiens_assembly38.dict --minimum-contig-length 46709983 --output
sandbox/plots --output-prefix hcc1143_T_clean

Using GATK jar /data4/wheeler_shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar
Running:
    java -Dsamjdk.use_async_io_read_samtools=false -Dsamjdk.use_async_io_write_samtools=true
-Dsamjdk.use_async_io_write_tribble=false -Dsamjdk.compression_level=2 -jar
/data4/wheeler_shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar PlotDenoisedCopyRatios --standardized-copy-ratios
sandbox/hcc1143_T_clean.standardizedCR.tsv --denoised-copy-ratios sandbox/hcc1143_T_clean.denoisedCR.tsv --sequence-dictionary
/data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/Homo_sapiens_assembly38.dict --minimum-contig-length 46709983 --output
sandbox/plots --output-prefix hcc1143_T_clean
13:55:33.275 INFO  NativeLibraryLoader - Loading libgkl_compression.so from
jar:file:/data4/wheeler_shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar!/com/intel/gkl/native/libgkl_compression.so
13:55:33.435 INFO  PlotDenoisedCopyRatios - ------------------------------------------------------------
```

```
13:55:33.439 INFO  PlotDenoisedCopyRatios - The Genome Analysis Toolkit (GATK) v4.6.2.0
13:55:33.440 INFO  PlotDenoisedCopyRatios - For support and documentation go to https://software.broadinstitute.org/gatk/
13:55:33.440 INFO  PlotDenoisedCopyRatios - Executing as mcollins@compute01 on Linux v3.10.0-1160.88.1.el7.x86_64 amd64
13:55:33.440 INFO  PlotDenoisedCopyRatios - Java runtime: Java HotSpot(TM) 64-Bit Server VM v22.0.1+8-16
13:55:33.440 INFO  PlotDenoisedCopyRatios - Start Date/Time: 11 February 2026, 13:55:33 GMT
13:55:33.440 INFO  PlotDenoisedCopyRatios - ------------------------------------------------------------
13:55:33.440 INFO  PlotDenoisedCopyRatios - ------------------------------------------------------------
13:55:33.441 INFO  PlotDenoisedCopyRatios - HTSJDK Version: 4.2.0
13:55:33.442 INFO  PlotDenoisedCopyRatios - Picard Version: 3.4.0
13:55:33.442 INFO  PlotDenoisedCopyRatios - Built for Spark Version: 3.5.0
13:55:33.445 INFO  PlotDenoisedCopyRatios - HTSJDK Defaults.COMPRESSION_LEVEL : 2
13:55:33.446 INFO  PlotDenoisedCopyRatios - HTSJDK Defaults.USE_ASYNC_IO_READ_FOR_SAMTOOLS : false
13:55:33.446 INFO  PlotDenoisedCopyRatios - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_SAMTOOLS : true
13:55:33.446 INFO  PlotDenoisedCopyRatios - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_TRIBBLE : false
13:55:33.446 INFO  PlotDenoisedCopyRatios - Deflater: IntelDeflater
13:55:33.446 INFO  PlotDenoisedCopyRatios - Inflater: IntelInflater
13:55:33.447 INFO  PlotDenoisedCopyRatios - GCS max retries/reopens: 20
13:55:33.447 INFO  PlotDenoisedCopyRatios - Requester pays: disabled
13:55:33.447 INFO  PlotDenoisedCopyRatios - Initializing engine
13:55:33.447 INFO  PlotDenoisedCopyRatios - Done initializing engine
13:55:33.451 INFO  PlotDenoisedCopyRatios - Reading and validating input files...
13:55:34.669 INFO  PlotDenoisedCopyRatios - Contigs above length threshold: {chr1=248956422, chr2=242193529, chr3=198295559,
chr4=190214555, chr5=181538259, chr6=170805979, chr7=159345973, chr8=145138636, chr9=138394717, chr10=133797422, chr11=135086622,
chr12=133275309, chr13=114364328, chr14=107043718, chr15=101991189, chr16=90338345, chr17=83257441, chr18=80373285, chr19=58617616,
chr20=64444167, chr21=46709983, chr22=50818468, chrX=156040895, chrY=57227415}
13:55:34.800 INFO  PlotDenoisedCopyRatios - Writing plots to /home/mcollins/sandbox/plots...
13:55:35.101 INFO  PlotDenoisedCopyRatios - Shutting down engine
[11 February 2026, 13:55:35 GMT] org.broadinstitute.hellbender.tools.copynumber.plotting.PlotDenoisedCopyRatios done. Elapsed time:
0.03 minutes.
Runtime.totalMemory()=331350016
org.broadinstitute.hellbender.utils.R.RScriptExecutorException:
Rscript exited with 1
Command Line: Rscript -e tempLibDir =
'/tmp/Rlib.17484945514140798493';source('/tmp/CNVPlottingLibrary.3041481449122730422.R');source('/tmp/PlotDenoisedCopyRatios.1346387
1478788987874.R'); --sample_name=HCC1143_tumor
```

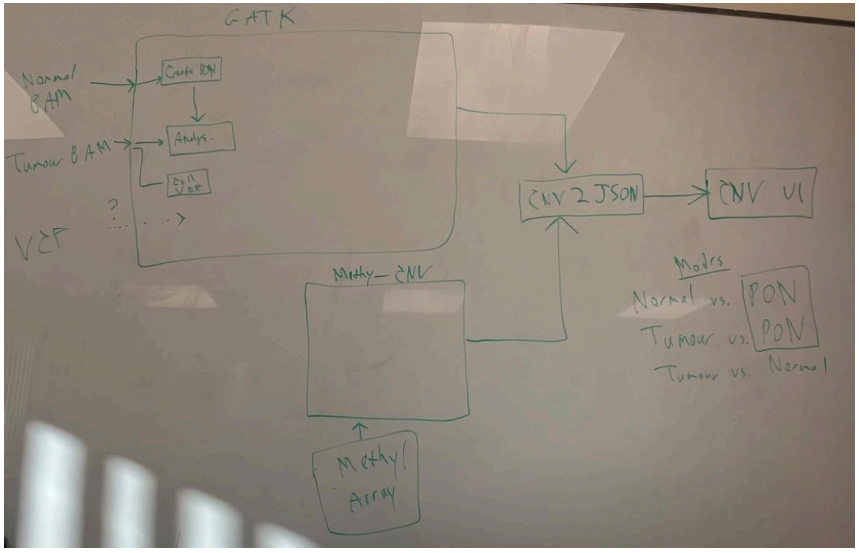
```
--standardized_copy_ratios_file=/home/mcollins/sandbox/hcc1143_T_clean.standardizedCR.tsv
--denoised_copy_ratios_file=/home/mcollins/sandbox/hcc1143_T_clean.denoisedCR.tsv
--contig_names=chr1CONTIG_DELIMITERchr2CONTIG_DELIMITERchr3CONTIG_DELIMITERchr4CONTIG_DELIMITERchr5CONTIG_DELIMITERchr6CONTIG_DELIMI
TERchr7CONTIG_DELIMITERchr8CONTIG_DELIMITERchr9CONTIG_DELIMITERchr10CONTIG_DELIMITERchr11CONTIG_DELIMITERchr12CONTIG_DELIMITERchr13C
ONTIG_DELIMITERchr14CONTIG_DELIMITERchr15CONTIG_DELIMITERchr16CONTIG_DELIMITERchr17CONTIG_DELIMITERchr18CONTIG_DELIMITERchr19CONTIG_
DELIMITERchr20CONTIG_DELIMITERchr21CONTIG_DELIMITERchr22CONTIG_DELIMITERchrXCONTIG_DELIMITERchrY
--contig_lengths=248956422CONTIG_DELIMITER242193529CONTIG_DELIMITER198295559CONTIG_DELIMITER190214555CONTIG_DELIMITER181538259CONTIG
_DELIMITER170805979CONTIG_DELIMITER159345973CONTIG_DELIMITER145138636CONTIG_DELIMITER138394717CONTIG_DELIMITER133797422CONTIG_DELIMI
TER135086622CONTIG_DELIMITER133275309CONTIG_DELIMITER114364328CONTIG_DELIMITER107043718CONTIG_DELIMITER101991189CONTIG_DELIMITER9033
8345CONTIG_DELIMITER83257441CONTIG_DELIMITER80373285CONTIG_DELIMITER58617616CONTIG_DELIMITER64444167CONTIG_DELIMITER46709983CONTIG_D
ELIMITER50818468CONTIG_DELIMITER156040895CONTIG_DELIMITER57227415 --maximum_copy_ratio=4.0 --point_size_copy_ratio=0.2
--output_dir=/home/mcollins/sandbox/plots/ --output_prefix=hcc1143_T_clean
Stdout: WARNING: unknown option '--sample_name=HCC1143_tumor'
WARNING: unknown option '--standardized_copy_ratios_file=/home/mcollins/sandbox/hcc1143_T_clean.standardizedCR.tsv'
WARNING: unknown option '--denoised_copy_ratios_file=/home/mcollins/sandbox/hcc1143_T_clean.denoisedCR.tsv'
WARNING: unknown option
'--contig_names=chr1CONTIG_DELIMITERchr2CONTIG_DELIMITERchr3CONTIG_DELIMITERchr4CONTIG_DELIMITERchr5CONTIG_DELIMITERchr6CONTIG_DELIM
ITERchr7CONTIG_DELIMITERchr8CONTIG_DELIMITERchr9CONTIG_DELIMITERchr10CONTIG_DELIMITERchr11CONTIG_DELIMITERchr12CONTIG_DELIMITERchr13
CONTIG_DELIMITERchr14CONTIG_DELIMITERchr15CONTIG_DELIMITERchr16CONTIG_DELIMITERchr17CONTIG_DELIMITERchr18CONTIG_DELIMITERchr19CONTIG
_DELIMITERchr20CONTIG_DELIMITERchr21CONTIG_DELIMITERchr22CONTIG_DELIMITERchrXCONTIG_DELIMITERchrY'
WARNING: unknown option
'--contig_lengths=248956422CONTIG_DELIMITER242193529CONTIG_DELIMITER198295559CONTIG_DELIMITER190214555CONTIG_DELIMITER181538259CONTI
G_DELIMITER170805979CONTIG_DELIMITER159345973CONTIG_DELIMITER145138636CONTIG_DELIMITER138394717CONTIG_DELIMITER133797422CONTIG_DELIM
ITER135086622CONTIG_DELIMITER133275309CONTIG_DELIMITER114364328CONTIG_DELIMITER107043718CONTIG_DELIMITER101991189CONTIG_DELIMITER903
38345CONTIG_DELIMITER83257441CONTIG_DELIMITER80373285CONTIG_DELIMITER58617616CONTIG_DELIMITER64444167CONTIG_DELIMITER46709983CONTIG_
DELIMITER50818468CONTIG_DELIMITER156040895CONTIG_DELIMITER57227415'
WARNING: unknown option '--maximum_copy_ratio=4.0'
WARNING: unknown option '--point_size_copy_ratio=0.2'
WARNING: unknown option '--output_dir=/home/mcollins/sandbox/plots/'
WARNING: unknown option '--output_prefix=hcc1143_T_clean'
Stderr: Error in library(optparse) : there is no package called ‘optparse’
Calls: source -> withVisible -> eval -> eval -> library
       at org.broadinstitute.hellbender.utils.R.RScriptExecutor.getScriptException(RScriptExecutor.java:79)
       at org.broadinstitute.hellbender.utils.R.RScriptExecutor.getScriptException(RScriptExecutor.java:18)
       at org.broadinstitute.hellbender.utils.runtime.ScriptExecutor.executeCuratedArgs(ScriptExecutor.java:112)
```

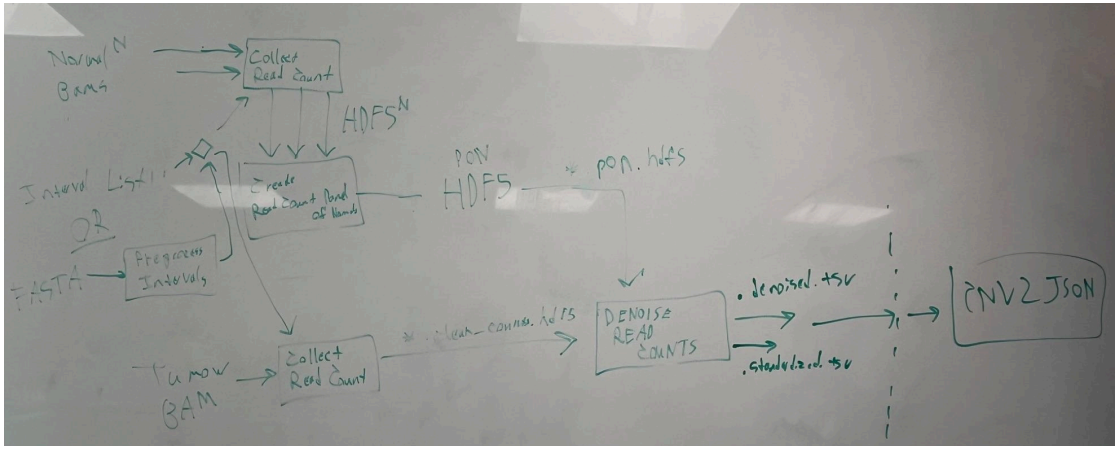
```
	at org.broadinstitute.hellbender.utils.R.RScriptExecutor.exec(RScriptExecutor.java:125)
	at
org.broadinstitute.hellbender.tools.copynumber.plotting.PlotDenoisedCopyRatios.writeDenoisingPlots(PlotDenoisedCopyRatios.java:217)
	at org.broadinstitute.hellbender.tools.copynumber.plotting.PlotDenoisedCopyRatios.doWork(PlotDenoisedCopyRatios.java:174)
	at org.broadinstitute.hellbender.cmdline.CommandLineProgram.runTool(CommandLineProgram.java:150)
	at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMainPostParseArgs(CommandLineProgram.java:203)
	at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMain(CommandLineProgram.java:222)
	at org.broadinstitute.hellbender.Main.runCommandLineProgram(Main.java:166)
	at org.broadinstitute.hellbender.Main.mainEntry(Main.java:209)
	at org.broadinstitute.hellbender.Main.main(Main.java:306)
```

February 13th

To do’s

- Make flow chart
- Do GATK tutorial on Aurora data: 1 normal and 1 tumour, compare the plots
- Eventually make one command to create a panel of normal file (HDF5 format), using python
- Make an Aurora PON





Flow Chart

- Different modes:
  - normal vs PON
  - Tumour vs PON

    - Tumour vs normal
- Standardised and denoised files go into JSON

GitHub Repository
Username: meriemacollins
Password: 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
Expiry: Aug 31st 2026

Two issues to make a note of on GitHub:

- `A USER ERROR has occurred: Badly formed genome unclippedLoc: Query interval "targets_C.interval_list" is not valid for this input.`
  -L
  PreprocessIntervals: More important for exome sequencing
- `Stderr: Error in library(optparse) : there is no package called ‘optparse’`
  Need optparse R package to create plots

Command to copy file
(main) [mcollins@lugh sandbox]$ cp hcc1143_T_clean.standardizedCR.tsv /data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/sandbox/

GATK on real data

- Use PON that you already have
    - Normal blood
    - Primary tumour
    - Metastatic tumour
- Use one of these instead of tumor.bam file
- Normal sample and either one of the tumours
- Tumour should look like it has copy number variants
- Make the plots

Later make an Aurora PON

- To get real outputs

Use python to create a PON from multiple normal bams using one command instead of 11

February 14th

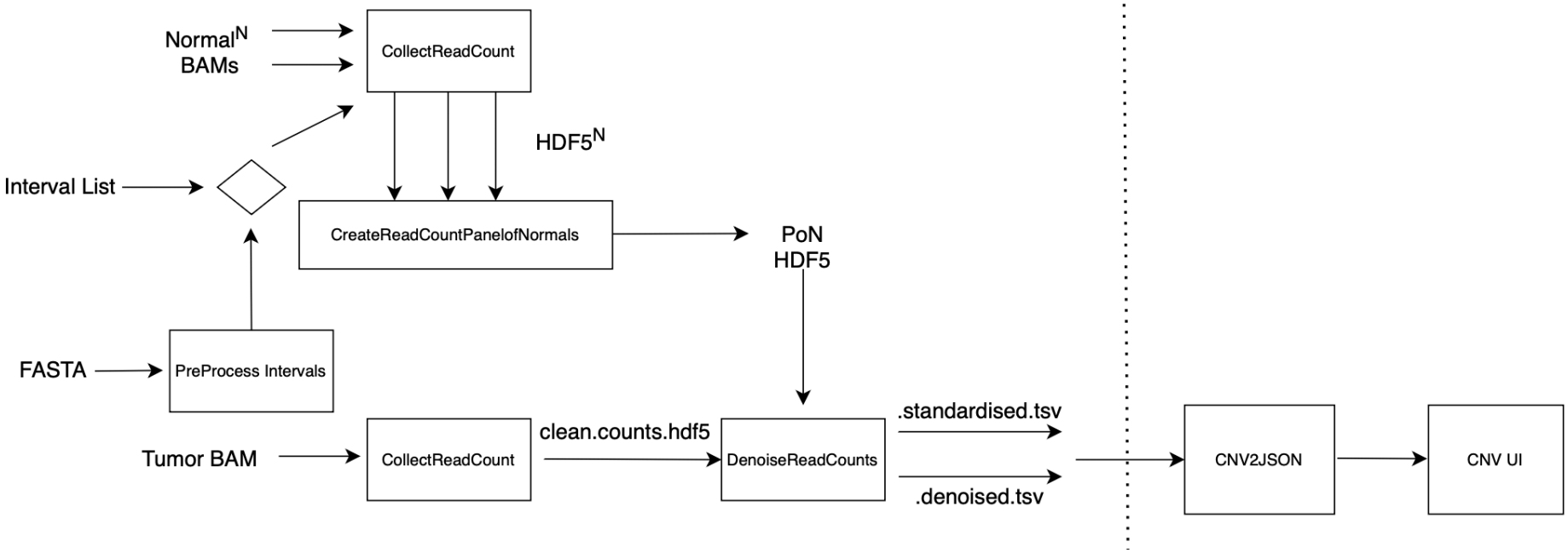
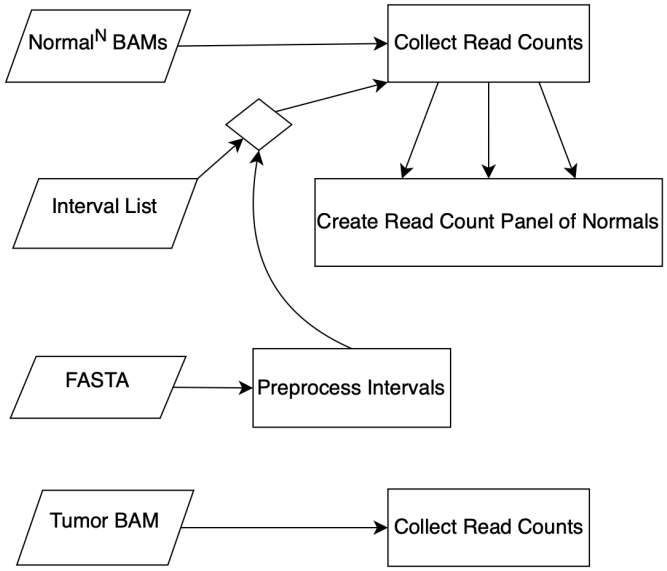
GATK on real data

```
salloc -n1 -N1 -p MSC
squeue -u mcollins
(main) [mcollins@lugh ~]$ ssh compute01
(main) [mcollins@compute01 ~]$ module load R
(main) [mcollins@compute01 ~]$ which R
/opt/ohpc/pub/libs/gnu7/R/3.5.0/bin/R
(main) [mcollins@compute01 Aurora_US]$ cd bams
(main) [mcollins@compute01 bams]$ ls
AG12-NB1.bam      AG12-TTM1.bam      AG12-TTP1.bam      cart_prj42634_202602030957.krt
AG12-NB1.bam.bai  AG12-TTM1.bam.bai  AG12-TTP1.bam.bai
```

[NB normal blood TP primary tumour TM metastatic tumour]

```
(main) [mcollins@compute01 ~]$ mkdir sandbox2

(main) [mcollins@compute01 ~]$ gatk PreprocessIntervals -R
/data4/wheeler_shared/reference/Broad/Homo_sapiens_assembly38_noALT_noHLA_noDecoy.fasta.gz --bin-length 0 --interval-merging-rule
OVERLAPPING_ONLY -O sandbox2/targets_C.preprocessed.interval_list

(main) [mcollins@compute01 ~] gatk CollectReadCounts -I /data4/wheeler_shared/data_files/Aurora_US/bams/AG12-TTM1.bam -L
sandbox2/targets_C.preprocessed.interval_list --interval-merging-rule OVERLAPPING_ONLY -O sandbox2/mtumor.counts.hdf5

(main) [mcollins@compute01 ~] gatk --java-options "-Xmx12g" DenoiseReadCounts -I mtumor.counts.hdf5 --count-panel-of-normals
/data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/cnvponC.pon.hdf5 --standardized-copy-ratios mtumor.standardizedCR.tsv
--denoised-copy-ratios mtumor.denoisedCR.tsv

(main) [mcollins@compute01 sandbox2]$ gatk --java-options "-Xmx12g" DenoiseReadCounts -I mtumor.counts.hdf5 --count-panel-of-normals
/data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/cnvponC.pon.hdf5 --standardized-copy-ratios mtumor.standardizedCR.tsv
--denoised-copy-ratios mtumor.denoisedCR.tsv
Using GATK jar /data4/wheeler_shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar
Running:
    java -Dsamjdk.use_async_io_read_samtools=false -Dsamjdk.use_async_io_write_samtools=true
-Dsamjdk.use_async_io_write_tribble=false -Dsamjdk.compression_level=2 -Xmx12g -jar
```

```
/data4/wheeler_shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar DenoiseReadCounts -I mtumor.counts.hdf5
--count-panel-of-normals /data4/wheeler_shared/data_files/GATK_Tutorial/tutorial_11682/cnvponC.pon.hdf5 --standardized-copy-ratios
mtumor.standardizedCR.tsv --denoised-copy-ratios mtumor.denoisedCR.tsv
19:55:30.397 INFO  NativeLibraryLoader - Loading libgkl_compression.so from
jar:file:/data4/wheeler_shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar!/com/intel/gkl/native/libgkl_compression.so
19:55:30.562 INFO  DenoiseReadCounts - ------------------------------------------------------------
19:55:30.566 INFO  DenoiseReadCounts - The Genome Analysis Toolkit (GATK) v4.6.2.0
19:55:30.566 INFO  DenoiseReadCounts - For support and documentation go to https://software.broadinstitute.org/gatk/
19:55:30.566 INFO  DenoiseReadCounts - Executing as mcollins@compute01 on Linux v3.10.0-1160.88.1.el7.x86_64 amd64
19:55:30.566 INFO  DenoiseReadCounts - Java runtime: Java HotSpot(TM) 64-Bit Server VM v22.0.1+8-16
19:55:30.567 INFO  DenoiseReadCounts - Start Date/Time: 14 February 2026, 19:55:30 GMT
19:55:30.567 INFO  DenoiseReadCounts - ------------------------------------------------------------
19:55:30.567 INFO  DenoiseReadCounts - ------------------------------------------------------------
19:55:30.568 INFO  DenoiseReadCounts - HTSJDK Version: 4.2.0
19:55:30.568 INFO  DenoiseReadCounts - Picard Version: 3.4.0
19:55:30.568 INFO  DenoiseReadCounts - Built for Spark Version: 3.5.0
19:55:30.572 INFO  DenoiseReadCounts - HTSJDK Defaults.COMPRESSION_LEVEL : 2
19:55:30.573 INFO  DenoiseReadCounts - HTSJDK Defaults.USE_ASYNC_IO_READ_FOR_SAMTOOLS : false
19:55:30.573 INFO  DenoiseReadCounts - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_SAMTOOLS : true
19:55:30.573 INFO  DenoiseReadCounts - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_TRIBBLE : false
19:55:30.574 INFO  DenoiseReadCounts - Deflater: IntelDeflater
19:55:30.574 INFO  DenoiseReadCounts - Inflater: IntelInflater
19:55:30.574 INFO  DenoiseReadCounts - GCS max retries/reopens: 20
19:55:30.574 INFO  DenoiseReadCounts - Requester pays: disabled
19:55:30.575 INFO  DenoiseReadCounts - Initializing engine
19:55:30.575 INFO  DenoiseReadCounts - Done initializing engine
19:55:30.578 INFO  HDF5Library - Trying to load HDF5 library from:
       jar:file:/data4/wheeler_shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar!/org/broadinstitute/hdf5/libjhdf5.2.11.0.so
19:55:30.613 INFO  H5 - HDF5 library:
19:55:30.614 INFO  H5 -  successfully loaded.
19:55:30.620 INFO  DenoiseReadCounts - Reading read-counts file (mtumor.counts.hdf5)...
19:55:30.830 INFO  SVDDenoisingUtils - Validating sample intervals against original intervals used to build panel of normals...
19:55:30.884 INFO  DenoiseReadCounts - Shutting down engine
[14 February 2026, 19:55:30 GMT] org.broadinstitute.hellbender.tools.copynumber.DenoiseReadCounts done. Elapsed time: 0.01 minutes.
Runtime.totalMemory()=192937984
```

```
java.lang.IllegalArgumentException: Sample intervals must be identical to the original intervals used to build the panel of normals.
	at org.broadinstitute.hellbender.utils.Utils.validateArg(Utils.java:798)
	at org.broadinstitute.hellbender.tools.copynumber.denoising.SVDDenoisingUtils.denoise(SVDDenoisingUtils.java:119)
	at 
org.broadinstitute.hellbender.tools.copynumber.denoising.SVDReadCountPanelOfNormals.denoise(SVDReadCountPanelOfNormals.java:88)
	at org.broadinstitute.hellbender.tools.copynumber.DenoiseReadCounts.doWork(DenoiseReadCounts.java:204)
	at org.broadinstitute.hellbender.cmdline.CommandLineProgram.runTool(CommandLineProgram.java:150)
	at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMainPostParseArgs(CommandLineProgram.java:203)
	at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMain(CommandLineProgram.java:222)
	at org.broadinstitute.hellbender.Main.runCommandLineProgram(Main.java:166)
	at org.broadinstitute.hellbender.Main.mainEntry(Main.java:209)
	at org.broadinstitute.hellbender.Main.main(Main.java:306)
```

Flow Chart Practice